

1 A novel machine learning-assisted clinical diagnosis support model for early
2 identification of pancreatic injury in blunt abdominal trauma patients at the
3 bedside: A cross-national study.

4 Running title: Identification for pancreatic injury in abdominal trauma

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45 **Abstract**

46 **Background:** There was still a serious delay in early identification of pancreatic
47 injury in blunt abdominal trauma patients. We developed a novel machine learning-
48 assisted clinical diagnosis support model for early identification of pancreatic injury
49 in blunt abdominal trauma patients at the bedside.

50 **Methods:** Retrospective analysis of a large ICU database (MIMIC-IV) for model
51 development and internal validation, and prospective outer validation based on a
52 cross-national dataset. Three models (PI-12, PI-12-2 and PI-24) were developed using
53 the logistic regression and compared according to different time windows for
54 observing the best model performance. The primary outcome was early identification
55 of pancreatic injury at any grades in blunt abdominal trauma patients at the bedside
56 during the first 24 hours after hospital admission. The developed model was further
57 evaluated in predicting the in-hospital and in-ICU mortality.

58 **Results:** The incidence of pancreatic injury was 5.56% (n=18) and 6.06% (n=6) in the
59 development (MIMIC-IV: 2008-2013, n= 324) and internal validation (MIMIC-IV:
60 2014-2019, n= 99) cohorts respectively. Internal validation cohort showed good
61 discrimination with AUCs of 0.73 (95% CI 0.54-0.92), 0.70 (95% CI 0.55-0.85) and
62 0.84 (95% CI 0.71-0.96) for PI-12, PI-12-2 and PI-24, respectively. PI-24 also
63 performed well with AUC of 0.82 (95% CI 0.65-0.98), specificity of 0.97 (95% CI
64 0.91-1.00) and PPV of 0.67 (95% CI 0.00-1.00) in the external validation cohort.

65 **Conclusions:** Pancreatic injury in blunt abdominal trauma could be early identified
66 with machine learning-assisted model based on routinely clinical information at the

67 bedside. The validated model is available at:

68 https://apoet.shinyapps.io/pancreatic_injury_2022_v2/.

69 **Keywords:** Pancreatic injury; Clinical diagnosis support model; Machine learning;

70 Abdominal trauma.

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83 Introduction

84 Although pancreatic trauma is relatively rare, accounting for only 0.2% ~ 12% of
85 blunt abdominal trauma, studies have reported that its related mortality was as high as
86 9% ~ 34%, which has become a major killer endangering human health[1]. Despite
87 the continuous improvement of the diagnosis and treatment strategies for pancreatic
88 trauma, there is still a serious delay in early identification due to the simultaneous
89 combination of other organ injuries, and the early symptoms, vital signs and
90 examination results are hidden and non-specific[2]. This is one of the most factors for
91 the high mortality rate.

92 The symptoms and vital signs of patients combined with the sequential dynamic
93 changes of serum amylase and lipase are still an important basis for identifying
94 pancreatic injury at an early phase in patients who have experienced blunt abdominal
95 trauma [3]. However, the specificity and sensitivity in the early identification of
96 pancreatic injury are poor. It has been reported that 14 ~ 80% of the patients suffering
97 from blunt injury in pancreas have increased. Even if the pancreas is complete
98 transection, 30 ~ 35% of patients still have normal serum amylase and lipase in the
99 early stage[4]. Some researchers have reported that the sensitivity of CT in early
100 recognition of pancreatic trauma is varies, ranging from 28% to 85%, especially for
101 the injury of the main pancreatic duct. Even with the application of high-resolution
102 CT imaging technology, the highest sensitivity is only 54%[5]. It has been reported
103 that magnetic resonance cholangiopancreatography (MRCP) and endoscopic
104 retrograde cholangiopancreatography (ERCP) not only have a high diagnostic rate for

pancreatic trauma, but also have a high sensitivity to the injury of the main pancreatic duct. They are effective methods for the identification and grading of pancreatic trauma recommended in clinic[6]. However, the application of these two methods in the early identification still has time-effect limitation, which could not be used at the bedside and limited the wide application [7].

At present, it is very difficult and challenging to evaluate and identifying pancreatic injury at an early phase in patients who have experienced blunt abdominal trauma based on the hidden and non-specific clinical information at the bedside. The studies on early identification and prediction model for critical illness based on machine learning has been in depth, and has shown strong ability to solve complex clinical problems. However, the machine learning-assisted clinical diagnosis support model for early identification of pancreatic injury in blunt abdominal trauma patients is still limited[8-12].

Our research aimed to develop a machine learning-assisted clinical diagnosis support model for identifying pancreatic injury at an early phase in patients who have experienced blunt abdominal trauma at the bedside by using a sufficient information from ICU database containing the background of general population. The developed model would be further external validated in another cross-national dataset.

Methods

Ethical statement

This research strictly abided by the Declaration of Helsinki (as revised in 2013) and

obtained approval from the ethics committee of the General Hospital of the PLA (No. S2020-418-01). Retrospective data and clinical information were used in this research. The procedure of collecting medical record was safe for patients and passive. No informed consent was needed because there was no research on human subject included in this study. Laboratory for Computational Physiology from the Massachusetts Institute of Technology (MIT-LCP) was responsible for maintaining the databases and provided approval to further analyze the MIMIC-IV.

Study design and cohorts

The models for prediction were established. The internal verification of the models was conducted by analyzing the retrospective data from a database containing single-center ICU data (MIMIC-IV,v0.4) including 76540 ICU cases from 2008 to 2019 in six ICU departments in a teaching hospital (Beth Israel Deaconess Medical Center, BIDMC; Boston, Massachusetts)[13-15]. MIMIC-IV was divided into development cohort (2008–2013) and validation cohort (2014–2019) for model establishment and internal verification. External verification was conducted in a prospective dataset, which included 42 blunt abdominal trauma patients admitted to the Emergency Department in the First Medical Center, Chinese People's Liberation Army General Hospital (PLAGH) in Beijing during 2021 in China. Both datasets included most routine clinical information such as laboratory tests, vital signs, demographics, etc. This study included the data of all the patients from MIMIC-IV but only the data of the cases in first hospital were applied [16]. The following items were inclusion criteria: 1) age ≥ 15 ; 2) hospitalization period ≥ 24 h (consecutive); 3) The primary

diagnosis contained "Blunt abdominal trauma, closed". If the data were insufficient during the first 24 hours following admission, the data were excluded.

Criteria of pancreatic trauma

The criteria of pancreatic trauma was according to the American Association for the Surgery of Trauma Organ Injury Scale (AAST-OIS)[17]. The pancreatic injury at any grade was identified according to the International Classification of Diseases, 9th or 10th Revision, Clinical Modification (ICD-9-CM or ICD-10-CM) in both two datasets.

Outcomes

In this study, the primary outcome was the early identification of pancreatic injury at any grades in blunt abdominal trauma patients at the bedside during the first 24 hours after hospital admission. The second outcome was predicting the in-hospital and in-ICU mortality.

Clinical models

We established 3 models for every model development mission according to the clinical data obtained at the bedside according to series time windows. The details of three models were elaborated as PI-12, PI-12-2 and PI-24:

- 1) PI-12: using only clinical data during the first 12 hours after hospital admission.
- 2) PI-12-2: using only clinical data that adds to the above, data available during the second 12 hours after hospital admission.
- 3) PI-24: using only clinical data that adds to the above, data available during the first 24 hours after hospital admission.

Models were developed by employing logistic regression method. R version 3.5.1 (R

Foundation for Statistical Computing, Vienna, Austria) was used to conduct the logistic regression machine-learning algorithm [9].

The selection of variables

The clinical data of each patient related with pancreatic injury were used as candidate variables according to series time windows and the availability in the databases. In MIMIC-IV cohort, the possible variables were screened. Univariate and multivariate analyses in each model were used to determine the variables. The variables were determined if they were significantly related with pancreas injury with a statistical difference of 0.10 in the univariate and multivariate analyses. R version 3.5.1 was used to conduct the multivariate analysis.

Model establishment and verification

Model establishment was conducted by only using the development cohort. The stability and performance of the model were verified by using 10-fold cross-validation method and development cohort. The internal verification cohort was used to evaluate the performance of the models. Then, the best performed model was further verified by using the external cohort and further assessed for predicting the in-hospital and in-ICU mortality. R version 3.5.1 was used to conduct the aforementioned procedures.

Performance evaluation and statistical analysis

Mean (SD) was used to express the continuous data. or Student's t-test and Mann-

Whitney U test were used to determine the significance of the differences. Absolute (n) and relative (%) frequency was used to express the categorical data which were analyzed by employing Fisher's exact test or chi-square test. The logistic regression method was employed to assess the multivariate analysis. The difference was considered as statistically significant if two-sided $\alpha < 0.10$. The difference and confidence intervals (CIs) were calculated to make a comparison on the model performance in the same subgroup of patient.

To evaluate the performances of the established models, discrimination evaluation was reported. Discrimination is the predictive ability in discriminating negative or positive outcomes, which is often assessed and quantified by the receiver operator characteristic (ROC) curve and with the area under the ROC curve (AUC), respectively.

Other indicators for evaluating discriminability were used as well, such as specificity, sensitivity, negative predictive value (NPV) and positive predictive value (PPV) at the threshold of classification. The threshold of classification is the cutoff value of the predicted possibilities above the value that the outcome might be considered as 'positive'. The cutoff value selected in this study led to the highest specificity and sensitivity in a group of patients with similar features in the illustrated tables and was used as the default value in the calculator online. This cutoff value was equivalent to the optimal point on the ROC curve according to the maximum value in the Youden-index. R version 3.5.1 was used for all the statistical analyses.

Results

The population in this study: cohorts for development, external and internal verification

There were 423 blunt abdominal trauma patients of MIMIC-IV and 42 blunt abdominal trauma subjects of PLAGH included in this study based on the exclusion and inclusion criteria illustrated in Figure 1. The cohort for development (2008-2013: n= 324) did showed no difference from the internal verification cohort (2014-2019: n= 99) in medical and demographic characteristics or clinical outcomes (Table 1), except for the higher incidence of spleen injury (48.15% vs. 34.34%, $p < 0.05$), lower incidence of bladder injury (3.40% vs. 11.11%, $p < 0.05$) and lower hospital mortality rate (2.78% vs. 10.10%, $p < 0.05$). The incidence of pancreatic injury was 5.56% (n=18) and 6.06% (n=6) in the development and internal validation cohorts, respectively, while 69.14% (n=224) and 64.65% (n=64) was male. Liver injury was primarily diagnosed in 44.75% (n=145) and 47.47% (n=47) subjects in each cohort, respectively. In the development cohort, ICU and hospital stay duration was 4.23 (5.54) and 10.08 (11.49) days, respectively. Other detailed information on each cohort is demonstrated in Table 1 and Supplemental Table 1.

In the MIMIC-IV cohort, some clinical characteristics in the first 12h after admission, in the second 12h after admission, and the average in the first 24h after admission were significant different between pancreatic injury and non-pancreas injury in patients suffering from blunt abdominal trauma (Table 2 and Supplemental Table 2).

For the first 12h after admission, abdominal trauma patients who were combining

pancreatic injury had a lower levels of magnesium and bicarbonate, had a higher temperature and diastolic blood pressure. For the second 12h after admission, pancreatic injury patients had a lower blood pH than non-pancreatic injury patients. For the average level in the first 24h after admission, serum lipase was higher and activated partial thromboplastin time (APTT) was lower in pancreatic injury than other abdominal trauma patients.

Variables selection

Candidate variables, univariable association with pancreatic injury were listed in Supplemental Table 2-3. Multivariable analysis and selected PI-12, PI-12-2 and PI-24 predictors were shown in Table 2.

Development cohort model performance

The performance of the model was improved as the availability of the data was increased. The PI-12 resulted in AUCs of 0.66, increasing to respective 0.76 and 0.81 for PI-12-2 and PI-24. According to the maximum value of Youden index, the specificity was 0.52, 0.42 and 0.74 for PI-12-2, PI-12 and PI-24, respectively. Meanwhile, the PPV of PI-12-2, PI-12 and PI-24 were 0.12, 0.10 and 0.20, respectively. Other results on discrimination for the clinical models in the development cohort are shown in Table 3.

The performance of the internal and external verification cohort models

The performance of PI-12, PI-12-2 and PI-24 in the cohort for internal verification is illustrated in Table 3 and Figure 2A. Satisfactory discrimination was observed and AUCs were 0.73, 0.70 and 0.84 for PI-12, PI-12-2 and PI-24, respectively. The thresholds of classification in the cohort for development were 3.26%, 2.48% and 7.27% for PI-12, PI-12-2 and PI-24, respectively, indicating that the results were robust and NPV, sensitivities, PPV, and specificities were similar in the cohort for internal verification. PI-24 had the better AUC, specificity and PPV than other two models, which was considered as the final clinical diagnosis support model. PI-24 also performed well with AUC of 0.82, specificity of 0.97 and PPV of 0.67 in the external validation cohort according to the threshold of classification found in the cohort for development (Figure 2B).

Model performance of major clinical outcome

In the internal validation cohort, PI-24 performed well in predicting in-hospital and in-ICU mortality. AUCs were 0.72 (95% CI 0.46-0.98) and 0.71 (95% CI 0.28-1.00), respectively.

Online machine learning-assisted clinical diagnosis support tool

The fine-tuned model PI-24 could be publicly obtained from the website https://apoet.shinyapps.io/pancreatic_injury_2022_v2/

Discussion

Identifying pancreatic injury at an early phase in patients with blunt abdominal trauma directly affects the early management and long-term prognosis[18]. A number of studies have been reported that the delay in diagnosis (diagnosis time > 24 hours after injury) for early identification of pancreatic injury is closely related to the high incidence rate of severe complications and high mortality[19]. Since it has been highlighted that pancreatic injury has strong association with poor clinical outcomes due to the delay in early identification, we developed and validated a clinical diagnosis support model based on machine learning in this study. Identifying pancreatic injury at an early phase in patients with blunt abdominal trauma is associated with research and clinical practice. Patients with higher risk need additional attention and they may gain great benefits if treated at an early phase with effective management.

Currently, decision-making models according to machine learning for pancreatic injury were less investigated. Cheng-Shyuan Rau et al. established a decision tree model to predict the population having a higher risk in pancreatic injury in a Level I Trauma Center by using the cross-sectional retrospective analysis [20]. However, the trauma patients whose levels of amylase or lipase were not elevated at early stage after hospital admission were not included. Jake E. Krige et al. developed and validated a new risk prediction scale which was organ-specific: pancreatic injury mortality score (PIMS), established from 5 variables for hospitalization-related mortality after critical pancreatic trauma[21]. Although PIMS increases clinical risk prediction for survival, it could not be used for identifying pancreatic injury at an

early phase in blunt abdominal trauma.

Serum level of lipase has been well-accepted as the most essential indicator associated with pancreatic injury and increased serum lipase is more specific than amylase [22].

In contrast, the elevated serum level of amylase in predicting pancreatic injury was not usually reliable[23]. In this study, selected variables for PI-24 showed that lipase level in the second 12h after admission and the average level in the first 24h after admission have the statistically significant association with pancreatic injury. But it has been reported that such an association would be not strong for relying solely on lipase measurement which call for combining other variables [23].

APTT, a commonly used indicator in coagulation, plays an essential role in diagnosing clotting and/or a bleeding disease and assessing patients before surgery[24]. Levent Kaptanoglu et al. in current approach to liver trauma reported that APTT showed a foreseeable change on hospital admission[25]. Sunhua Huang et al. reported that increased APTT was found in Darlington's trauma-induced coagulopathy rat which was resulted from hemorrhagic shock[26]. In our results of the multivariable association with PI-24, the prolonged APTT level in the first 24h after admission showed less ratio in identification of pancreatic injury that may because it less likely to cause hemorrhage such as solid organ injury.

The metabolic acidosis was present in the majority of patients with critical illness, which was related to an increased mortality and morbidity [27]. The importance of lactate as a marker for shock in critically ill patients, especially trauma, has been well established[27, 28]. Many studies reported that there was a connection between

acute pancreatitis onset and low pH [29]. Patients suffering from pancreatic injury were at higher risk of presenting with shock, and evidence from clinical practice has shown that various shock states existed in ischemic pancreatitis [30]. In this study, based on multivariable analysis for PI-24, the lower blood pH, the higher lactate and higher anion gap showed a strong association with pancreatic injury when patients experienced blunt abdominal trauma. It suggests that pancreatic injury in blunt abdominal trauma may cause early disorder of metabolic acidosis-related parameters due to the potentially variety of shock states. In other hand, the metabolic acidosis-related parameters, such as lactate, blood pH and anion gap, may potentially as early markers differentiate pancreatic injury in blunt abdominal trauma.

There were some strengths in our research. First, we used a critical database of various populations who suffered from blunt abdominal trauma and the diagnosis support model has been developed based on the routinely clinical variables at the bedside within the first 24 hours, which would bring an earlier warning time window compared with other methods in the identification of the pancreatic injury. Second, in the patients with blunt abdominal trauma, our PI-24 model would be able to provide timely supportive information at the bedside, this model will allow comparison of the time-effect management quality provided for this sophisticated population. Third, the development and verification method was used, with robust model compared and chosen for prospect external verification using the latest real world clinical data. Fourth, we combined numerous clinical data that could be obtained routinely according to different time windows as candidate variables for selection, which were

demonstrated by univariate and multivariate association analysis. Fifth, the least number of variables was used in the model with the ideal sensitivity and specificity that could be obtained in most hospital, which improved the possibility of widely using the model in clinical practice. Sixth, the in-hospital and in-ICU mortality could also be predicted well using PI-24 which would be able to provide prognostic information at the early stage. Seventh, the metabolic acidosis-related parameters may potentially as the early markers differentiate pancreatic injury in blunt abdominal trauma. Last, the developed and validated PI-24 could be used as an online clinical diagnosis support tool, which provides a useful platform for future biomarker and pancreatic prediction studies and further validation at the bedside or even when resource were limited.

There were still some limitations in this study. First, dataset for development was single-center studies, and data from multi-center studies were needed in the future research. Second, some routinely clinical information, such as in-hospital surgery and trauma severity score were not reported as the unavailable data in the databases. Third, the sample size of the prospect outer validation dataset was relatively small and the incidence of the pancreatic injury was relatively high due to the influence of the COVID-19 that fewer blunt abdominal trauma patients could be admitted. Fourth, we used retrospective data in this research, which might result in a lack of certain information in comparison with data from prospective studies. Fifth, the modest performance of model of PPV might be due to imperfect categorization. Furthermore, before this model could be used in clinical practice, more steps are needed. Thus,

further external validation is required to be conducted using more independent data to demonstrate that early management for pancreatic injury induced by model outputs significantly change clinical outcomes, experiences of the patients, or care management.

Conclusions

A novel machine learning-assisted clinical diagnosis support model was developed for identifying pancreatic injury at an early phase in patients with blunt abdominal trauma at the bedside. We showed that the metabolic acidosis-related parameters may potentially as the early markers differentiate pancreatic injury in blunt abdominal trauma. These developed clinical diagnosis support model is available online as a tool to provide benefit for further studies, which can be obtained freely from the website: https://apoet.shinyapps.io/pancreatic_injury_2022_v2/

Declarations

Ethics approval and consent to participate

The study strictly abided by the Declaration of Helsinki (as revised in 2013) and obtained approval from the institutional ethics committee of the General Hospital of the PLA (No. S2020-418-01). Retrospective data and clinical information were used in this research. The procedure of collecting medical record was safe for patients and passive. No informed consent was needed because there was no research on human subject included in this study. Laboratory for Computational Physiology from the

Massachusetts Institute of Technology (MIT-LCP) was responsible for maintaining the databases and provided approval to further analyze the MIMIC-IV.

Consent for publication

Not applicable.

Availability of data and materials

The development and internal validation datasets generated and during the current study are available in the <https://pHysionet.org/>. The outer validation dataset only available after the permission of the corresponding author.

Competing interests

The authors declare that they have no competing interests.

Funding

There is no financial funding or interest to report.

Author's Contributions

The study was designed by CF, who was also responsible for data collection, analysis, and interpretation, and drafting the manuscript. XZ, BY and YT contributed in the data analysis and interpretation, who also conducted the online early warning calculator, developed the website, and wrote the manuscript. LM, LW, XZ and JW conducted the data interpretation and wrote the manuscript. LC and YY conducted the data interpretation and reviewed the manuscript.

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Supplemental Tables of Contents

Supplemental Table 1: Illustration of the development and outer validation cohort.

Supplemental Table 2: The candidate variables for the three clinical identification models; every model also contains all candidate variables of the rows above it.

Supplemental Table 3: The variables list of a statistically significant univariable association with pancreatic injury.

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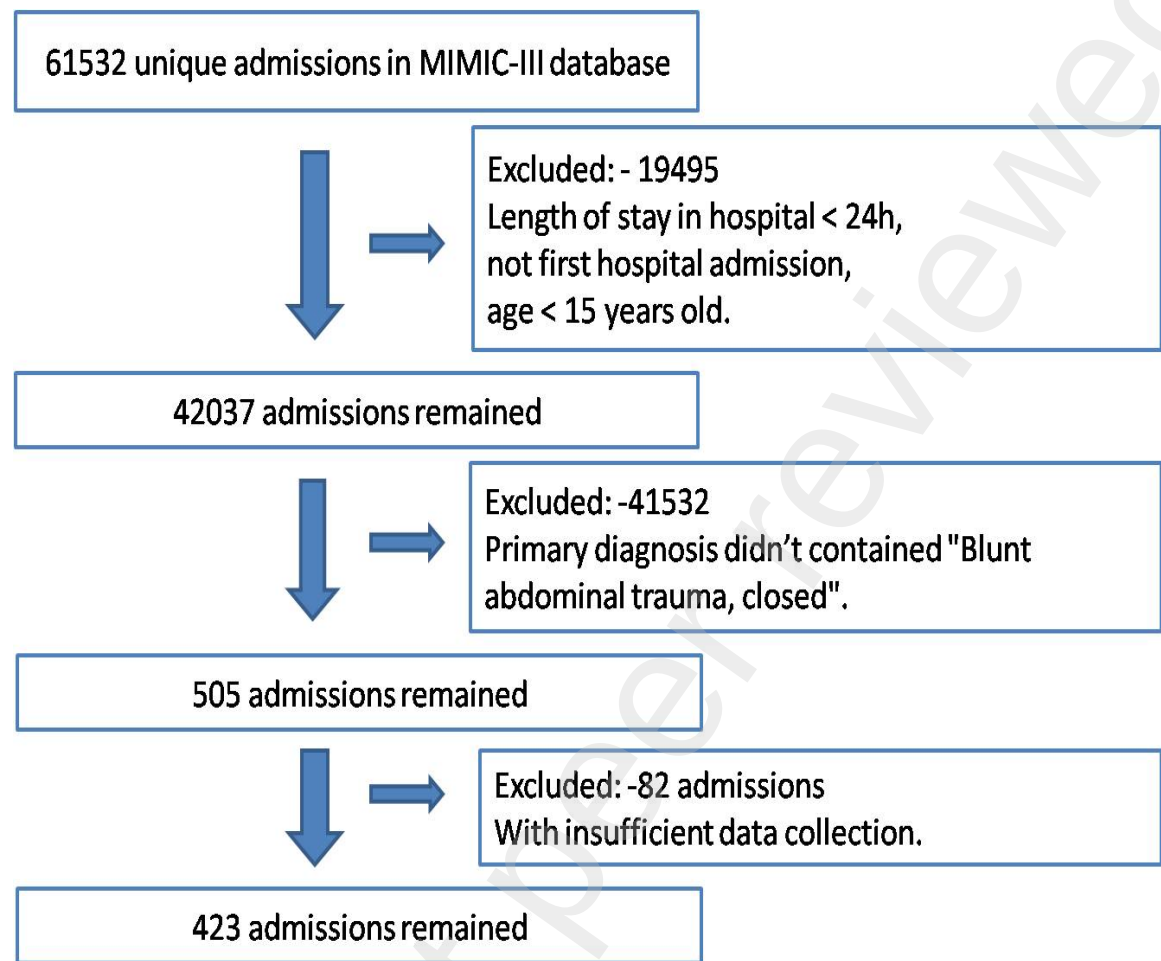
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Figure legends

Figure 1

Study cohort selection workflow of MIMIC-IV database based on the designed

516 inclusion and exclusion criteria.



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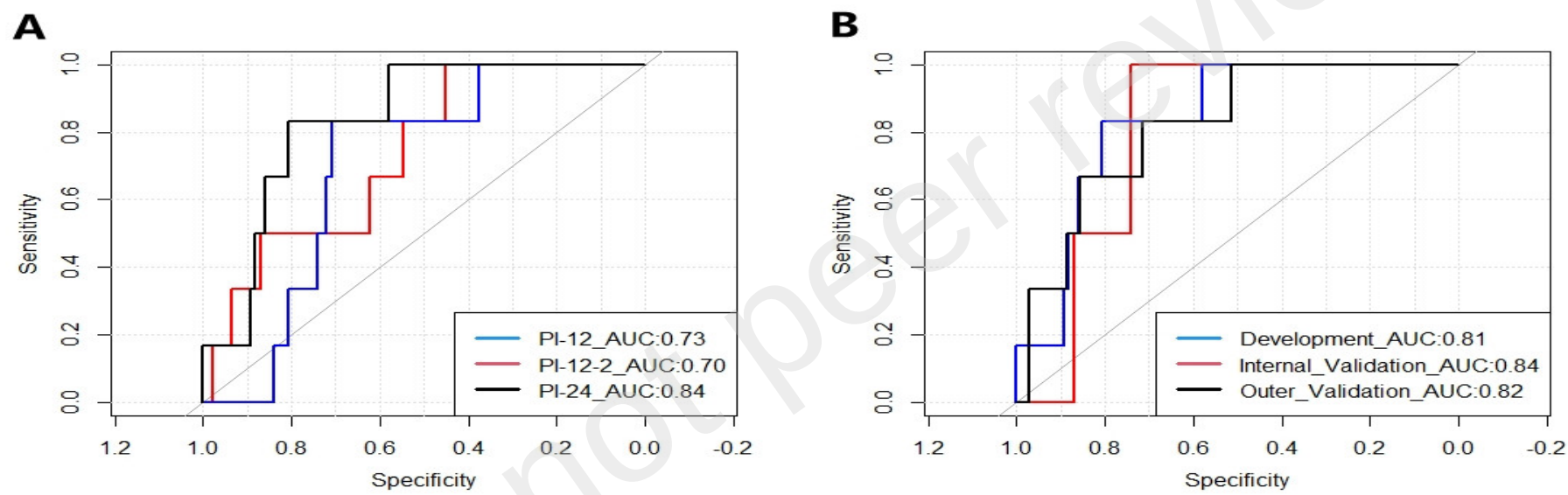
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525 Figure 2

526 Model performances of the three models in different cohorts.



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528 A: The ROC curves and AUCs of PI-12, PI-12-2 and PI-24 in internal validation cohort. B: The ROC curves and AUCs of PI-24 in development,
529 internal validation and outer validation cohorts.

530 ROC: receiver operator characteristic

531 AUC: the area under the ROC curve

532 **Table 1. Description of the development and internal validation cohort.**

Key Characteristics	Development cohort	Internal validation cohort	P-value
Demographic information:			
Final cohort (n)	324	99	-
The incidence of pancreatic injury (n(%))	18 (5.56%)	6 (6.06%)	1.00
The incidence of pancreatic duct injury (n(%))	6 (1.9%)	0 (0.00%)	0.38
Age, years (mean (s.d.))	43.84 (20.52)	46.39 (20.63)	0.19
Weight, kg (mean (s.d.))	82.61 (22.83)	80.00 (22.67)	0.14
Male gender (n(%))	224 (69.14%)	64 (64.65%)	0.47
Primary diagnosis (n(%)):			
Liver injury	145(44.75%)	47 (47.47%)	0.72
Gallbladder injury	8(2.47%)	2(2.02%)	1.00

Spleen injury	156(48.15%)	34(34.34%)	<0.05
Stomach or intestine injury	29 (8.95%)	10 (10.10%)	0.88
Kidney injury	56(17.28%)	16 (16.16%)	0.91
Bladder injury	11(3.40%)	11 (11.11%)	<0.01

Clinical outcomes:

Length of ICU stay, days (mean (s.d.))	4.23 (5.54)	4.12 (5.01)	0.75
Length of hospital stay, days (mean (s.d.))	10.08 (11.49)	11.17 (12.01)	0.25
ICU mortality rate(n(%))	5 (1.54%)	4 (4.04%)	0.27
Hospital mortality rate(n(%))	9 (2.78%)	10 (10.10%)	<0.01

533 Continuous variables were expressed as the mean (SD).

534 Categorical variables are expressed as absolute (n) and relative (%) frequency.

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537 **Table 2. Selected variables list and multivariable association with pancreatic injury.**

Models	Key Characteristics	OR	95%CI	p-value (Wald's test)
PI-12	Temperature_12	2.49	1.08-5.76	<0.05
	Diastolic blood pressure_12	1.04	1.00-1.09	<0.10
	Bicarbonate_12	0.87	0.74-1.02	<0.10
	Magnesium_12	0.11	0.02-0.81	<0.05
PI-12-2	Temperature_12	3.44	1.44-8.22	<0.01
	Diastolic blood pressure_12	1.05	1.00-1.10	<0.05
	Magnesium_12	0.05	0.01-0.47	<0.01
	Blood pH_12_2	0.00	0.00-0.05	<0.01
PI-24	Blood pH_12_2	0.00	0.00-0.09	<0.05
	Lipase_12_2	0.9968	0.9936-1.00	<0.05

	Lactate_12_2	1.69	1.05-2.70	<0.05
	Anion gap_12_2	1.34	1.01-1.78	<0.05
	Lipase_24	1.0016	1.0003-1.0029	<0.01
	Activated partial thromboplastin time_24	0.76	0.63-0.92	<0.01

538 "_12": In the first 12h after admission; "_12_2": In the second 12h after admission; "_24": The average in the first 24h after admission.

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Table 3: Performance of clinical prediction models in development, internal validation and outer validation cohorts.

Models	PI-12		PI-12-2		PI-24		
Cohort	Development	Internal validation	Development	Internal validation	Development	Internal validation	Outer validation
AUC	0.66 (0.17-1.00)	0.73 (0.54-0.92)	0.76 (0.28-1.00)	0.70 (0.55-0.85)	0.81 (0.63-0.98)	0.84 (0.71-0.96)	0.82 (0.65-0.98)
Sensitivity	1.00 (0.50-1.00)	1.00 (1.00-1.00)	1.00 (0.50-1.00)	0.83 (0.50-1.00)	1.00 (1.00-1.00)	0.83 (0.50-1.00)	0.33 (0.00-0.67)
Specificity	0.42 (0.32-0.97)	0.38 (0.28-0.47)	0.52 (0.39-1.00)	0.53 (0.42-0.62)	0.74 (0.58-0.94)	0.73 (0.65-0.82)	0.97 (0.91-1.00)
Positive predictive	0.10 (0.09-0.68)	0.09 (0.08-0.11)	0.12 (0.10-1.00)	0.10 (0.06-0.14)	0.20 (0.13-0.50)	0.17 (0.10-0.24)	0.67 (0.00-1.00)

value							
Negative predictive value	1.00 (0.64-1.00)	1.00 (1.00-1.00)	1.00 (0.97-1.00)	0.98 (0.94-1.00)	1.00 (1.00-1.00)	0.99 (0.96-1.00)	0.89 (0.85-0.95)
Classification Threshold (%)	3.26%	3.26%	2.48%	2.48%	7.27%	7.27%	7.27%

547 AUC, area under the ROC curve; ROC, receiver operator characteristic.

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